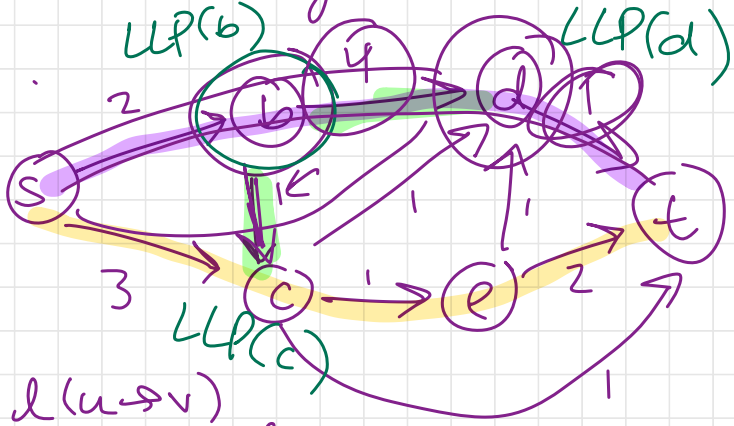
DAG DP - dynamic program

Given a DAG $G = (V, E)$ w/ edge lengths weights

Find a path in G from s to t w/ max total length. ← length of c



$2+4+1=7$

$3+1+2=6$

Let $l(u \to v)$ be length of $u \to v$.

$l(b \to c) = 1$

DP Alg

(1) Write the def. of the recursive subproblem you are using

→ $LLP(v)$ = length of longest path in G from v to t.

(1a) What input do you call your subproblem on to get overall answer?

$LLP(s)$ = length of longest path in G from s to t

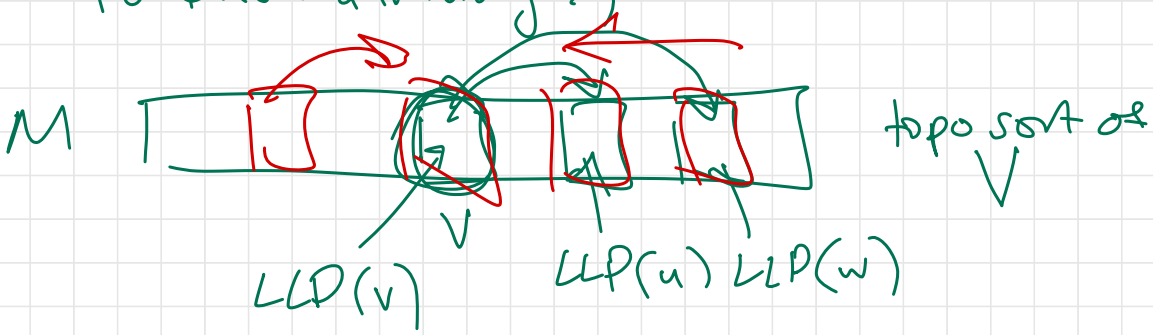
(2) Give a recurrence for your subproblem

$$LLP(v) = \begin{cases} 0 & \text{if } v = t \\ \max_{v \to w} \left(l(v \to w) + LLP(w) \right) & \text{otherwise} \end{cases}$$

choices: which edge out of v do I take?

(3) Describe the memoization structure.

(when I compute $LLP(v)$, what do I need to know already?)



fill in reverse topo order

(4) Write iterative alg

G global

LongestPath(s,t):

(1) compute a postorder of V $O(V+E)$ DFS

?

```
for node v in the postorder:
    if v = t:
        v.LLP = 0
    else:
        best = -∞
        for each edge v→w:
            if w.LLP + ℓ(v→w) > best:
                best = w.LLP + ℓ(v→w)
        v.LLP = best
return s.LLP
```

O(E)

(2) ?

Graph problems we can solve

- traversal / connectivity
  - does s reach t WFS
  - shortest path (#edges) from s to t — BFS ← breadth
  - give a path from s to t WFS
  - is G connected? WFS
  - how many CCs? WFS

  $O(V+E)$

- Best first search
  - priority queue
  - minimum spanning tree $O(V+E \log E)$
    - $l(u \to v)$
  - bottleneck / widest path
    - $\min(w(u \to v), u.width)$
  - shortest weighted path
    - $\min(w(u \to v) + u.dist)$

- directed graph connectivity
  - are s, t strongly connected $O(V+E)$
  - compute metagraph $O(V+E)$

- topo sort cycles
  - compute a topo sort $O(V+E)$
  - check for cycles $O(V+E)$

- DAG DP
  - more complex optimization